

WHAT IS LINEAR REGRESSION?

A linear regression is a data plot that graphs the linear relationship between an **independent** and a **dependent** variable(s).

It is typically used to visually show the **strength of the relationship**, and the dispersion of results.

E.g. to see test the strength of the relationship between amount of **ice cream eaten** and **obesity**.

Take the **independent** variable, the amount of ice cream, and relate it to the dependent variable, obesity, to see if there was a relationship.

MATHEMATICAL FORM

Linear Regression: Single Variable

$$\boxed{\hat{y}} = \beta_0 + \beta_1 \boxed{x} + \boxed{\epsilon}$$

Predicted output

Coefficients

Input

Error

Linear Regression: Multiple Variables

$$\boxed{\hat{y}} = \beta_0 + \beta_1 \boxed{x_1} + \dots + \beta_p \boxed{x_p} + \boxed{\epsilon}$$

WHAT DO WE USE LINEAR REGRESSION FOR?

The overall idea of linear regression is to examine 2 things:

- Does a set of predictor variables do a good job in predicting an outcome (dependent) variable?
- Which variables in particular are **significant predictors** of the outcome variable and in what way do they *—indicated by the magnitude and sign of the beta estimates—* impact the outcome variable?

KEY POINTS ...

- When selecting the model for the analysis, an important consideration is model fitting.
- Adding independent variables to a linear regression model will always increase the explained variance of the model (typically expressed as R^2).
- **overfitting** can occur by adding too many variables to the model, which **reduces** model generalizability.
- A simple model is usually **preferable** to a more complex model.
- Statistically, if a model includes a **large number of variables**, some of the variables will be statistically significant due to chance alone.

SIMPLE LINEAR REGRESSION

A college bookstore must order books 2 months before each semester starts. They believe that the number of books that will be sold for any particular course is related to the number of students registered for the course when the books are ordered.

They would like to develop a **linear regression** equation to help plan how many books to order.

From past records, the bookstore obtains the number of students registered, **X**, and the number of books actually sold for a course, **y** for 12 different semesters.

Semester	No of students	Books
1	36	31
2	28	29
3	35	34
4	39	35
5	30	29
6	30	30
7	31	30
8	38	38
9	36	34
10	38	33
11	29	29
12	26	26

WHAT IS THE ERROR TERM?

- An **error term** is a variable in a statistical or mathematical model, which is created when the model does not fully represent the actual relationship between the independent variables and the dependent variables.
- The **error term** is also known as the **residual**, **disturbance**, or **remainder** term.

WHAT IS THE ERROR TERM?

- Within a linear regression model tracking a stock's price over time, **the error term** is the difference between the **expected price at a particular time and the price that was actually observed**.
- In instances where the **price is exactly what was anticipated** at a particular time, the price will fall on the trend line and the **error term** will be **zero**.
- *Points that do not fall directly on the trend line exhibit the fact that the dependent variable, in this case, the price, is influenced by more than just the independent variable, representing the passage of time.*
- *The error term stands for any influence being exerted on the price variable, such as changes in market sentiment.*

ERROR CALCULATION

The residual is calculated as: $r_i = y_i - \hat{y}$

where

r_i = residual value

y_i = observed value for a given x value

$\hat{y}$ = predicted value for a given x value

- The magnitude of a typical residual can give us a sense of generally how close our estimates are.
- The smaller the residual standard deviation, the closer is the fit to the data.
- In effect, the smaller the residual standard deviation is compared to the sample standard deviation, the more predictive, or adequate, the model is.

linear equation is $\hat{y} = 1x + 2$, the residual for each observation can be found.

For the first set, the actual y value is 1, but the predicted y value given by the equation is $\hat{y} = 1(1) + 2 = 3$. The residual value is, therefore, $1 - 3 = -2$, a negative residual value

X	Y	$\hat{y}$	error	error ²
1	1	3	-2	4
2	4	4	0	0
3	6	5	1	1
4	7	6	1	1

Sum of squared residuals: 6

Number of residuals less 1: $4 - 1 = 3$

Residual standard deviation: $\sqrt{(6/3)} = \sqrt{2} \approx 1.4142$

LOSS FUNCTION

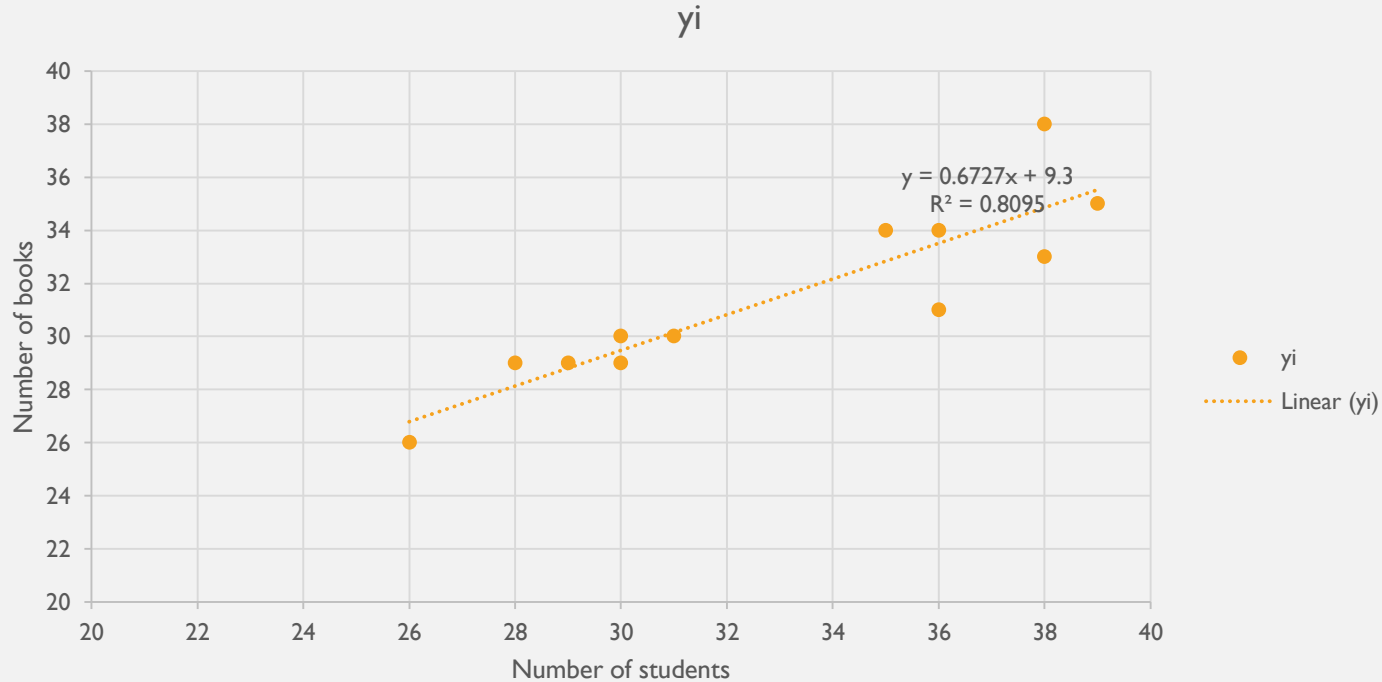
- Every machine learning problem is basically an optimization problem.
- The function that you want to optimize is usually called the loss function (or cost function).
- For the house price prediction example, after the model is trained, we are able to predict new house prices based on their features.
- For each house price we predict, denoted as $\hat{Y}_i$, and the actual house price Y_i we can calculate the loss by: $loss_i = (\hat{Y}_i - Y_i)^2$

$$L = \sum (\hat{Y}_i - Y_i)^2$$

- Which simply defines that our model's loss is the sum of distances between the house price we've predicted and the ground truth.
- This loss function, in particular, is called quadratic loss or least squares.
- We wish to minimize the loss function (L) as much as possible so the prediction will be as close as possible to the ground truth.

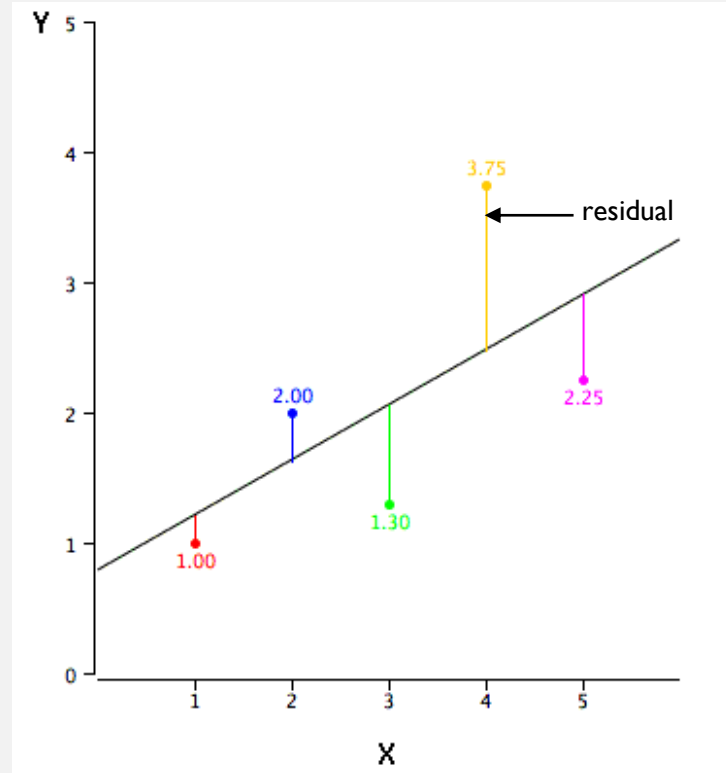
REGRESSION LINE

Obtain a scatter plot of the number of books sold versus the number of registered students.



COMPARING MODEL PREDICTIONS AGAINST REALITY

- $\text{Response} = (\text{Constant} + \text{Predictors}) + \text{Error}$
- Response/output is estimated for any given input or set of inputs
- Check these estimated outputs against the actual values
- The difference between the actual value and the model's estimate a **residual**.
- These residuals will play a significant role in judging the usefulness of a model.
- If the **residuals are small**, it implies that the model is a **good** estimator



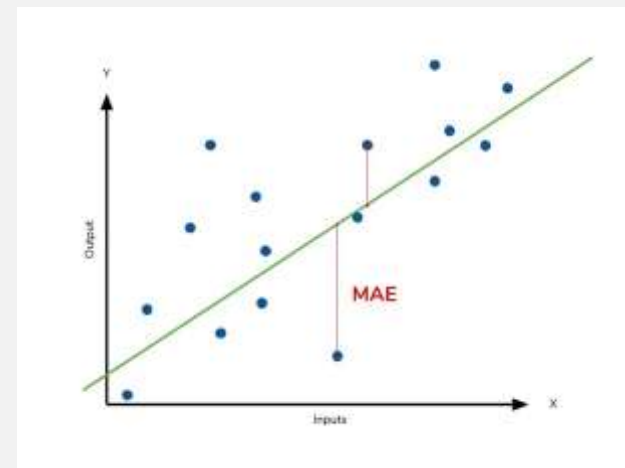
MEAN ABSOLUTE ERROR

- Calculate the **residual** for every data point, taking only the **absolute value** of each so that negative and positive residuals do not cancel out.
- Take the **average** of all these **residuals**.
- Effectively, MAE describes the typical **magnitude** of the **residuals**.

$$MAE = \frac{1}{n} \sum \left| y - \hat{y} \right|$$

Diagram illustrating the MAE formula components:

- $\frac{1}{n}$: Divide by the total number of data points
- $\sum$: Sum of
- y : Actual output value
- $\hat{y}$: Predicted output value
- $|y - \hat{y}|$: The absolute value of the residual



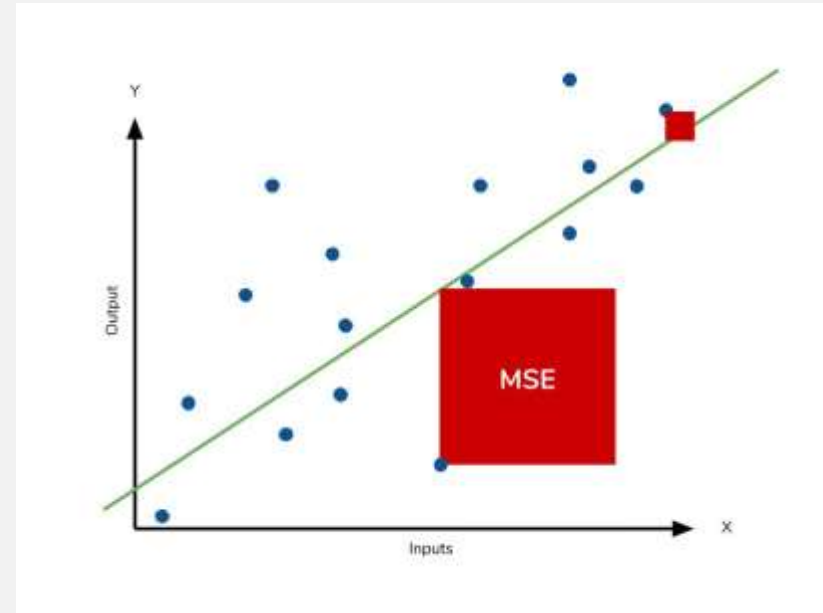
MEAN ABSOLUTE ERROR

- The MAE is also the most intuitive of the metrics
- Each residual contributes proportionally to the total amount of error, meaning that larger errors will contribute linearly to the overall error.
- A small MAE suggests the model is great at prediction, while a large MAE suggests that your model may have trouble in certain areas.
- A MAE of 0 means that your model is a perfect predictor of the outputs (but this will almost never happen).

MEAN SQUARE ERROR

- The mean square error (**MSE**) is just like the **MAE** but **squares the difference** before summing them all instead of using the absolute value.

$$MSE = \frac{1}{n} \sum \left(\underbrace{y - \hat{y}}_{\substack{\text{The square of the difference} \\ \text{between actual and} \\ \text{predicted}}} \right)^2$$



MEAN SQUARE ERROR

- Because we are squaring the difference, the MSE will almost always be bigger than the MAE.
 - For this reason, we cannot directly compare the MAE to the MSE.
- The effect of the square term in the MSE equation is most apparent with the presence of outliers in our data.
- Each residual in MAE contributes proportionally to the total error, the error grows quadratically in MSE.
- **Outliers** in our data will contribute to much higher total error in the **MSE** than they would the **MAE**.

ROOT MEAN SQUARED ERROR (RMSE)

- Square root of the **MSE**
- Since the **MSE** and **RMSE** both square the residual, they are similarly affected by outliers.
- The **RMSE** is analogous to the **standard deviation** (MSE to variance)
- Is a measure of how large your residuals are spread out.

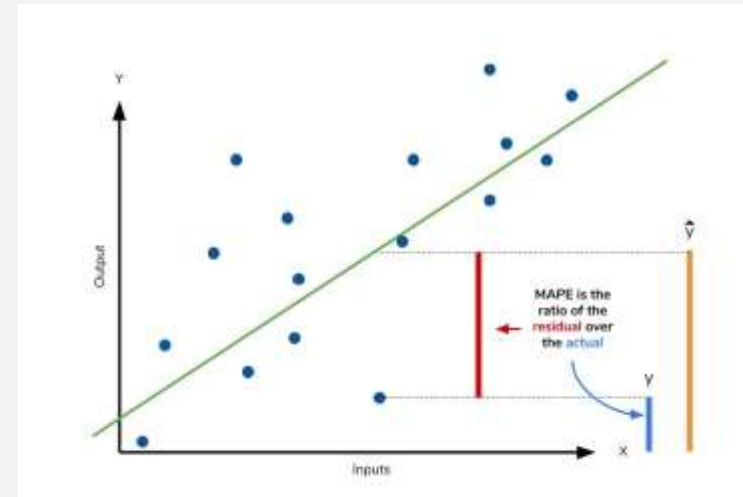
$$\text{MSE}(a) > \text{MSE}(b) \iff \text{RMSE}(a) > \text{RMSE}(b)$$

MEAN ABSOLUTE PERCENTAGE ERROR

- The mean absolute percentage error (MAPE) is the percentage equivalent of MAE.
- The equation looks just like that of MAE, but with adjustments to convert everything into percentages.
- Both **MAPE** and **MAE** are robust to the effects of outliers thanks to the use of absolute value.

$$MAPE = \frac{100\%}{n} \sum \left| \frac{\overbrace{y - \hat{y}}^{\text{The residual}}}{\underbrace{y}_{\text{Each residual is scaled against the actual value}}} \right|$$

Multiplying by 100% converts to percentage



MEAN ABSOLUTE PERCENTAGE ERROR

- MAPE's weaknesses actually stem from use of division operation.
- MAPE is **undefined** for data points where the value is 0.
- Similarly, the MAPE can grow unexpectedly large if the actual values are exceptionally small themselves.
- Finally, the MAPE is **biased towards predictions that are systematically less than the actual values** themselves. That is to say, MAPE will be lower when the prediction is lower than the actual compared to a prediction that is higher by the same amount.

$$MAPE = \frac{100\%}{n} \sum \left| \frac{y - \hat{y}}{y} \right|$$

Multiplying by 100% converts to percentage

The residual

Each residual is scaled against the actual value

$\hat{y}$ is smaller than the actual value

$n = 1 \quad \hat{y} = 10 \quad y = 20$

MAPE = 50%

$\hat{y}$ is greater than the actual value

$n = 1 \quad \hat{y} = 20 \quad y = 10$

MAPE = 100%

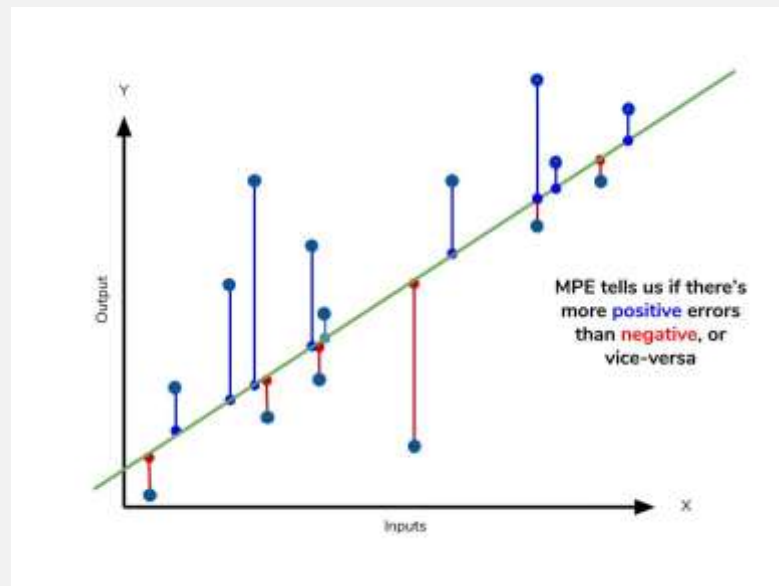
MEAN PERCENTAGE ERROR

- The mean percentage error (**MPE**) equation is exactly like that of **MAPE**.
- The only difference is that it **lacks the absolute** value operation.

$$MPE = \frac{100\%}{n} \sum \left(\frac{y - \hat{y}}{y} \right)$$

$$MAPE = \frac{100\%}{n} \sum \left| \frac{\overbrace{y - \hat{y}}^{\text{The residual}}}{\underbrace{y}_{\text{Each residual is scaled against the actual value}}} \right|$$

Multiplying by 100% converts to percentage



MEAN PERCENTAGE ERROR

- Since positive and negative errors will cancel out, we cannot make any statements about how well the model predictions perform overall.
- However, if there are more negative or positive errors, this bias will show up in the MPE.
- Unlike MAE and MAPE, MPE is useful to us because it allows us to see if our model **systematically underestimates** (more negative error) or **overestimates** (positive error).

RESIDUAL PLOT

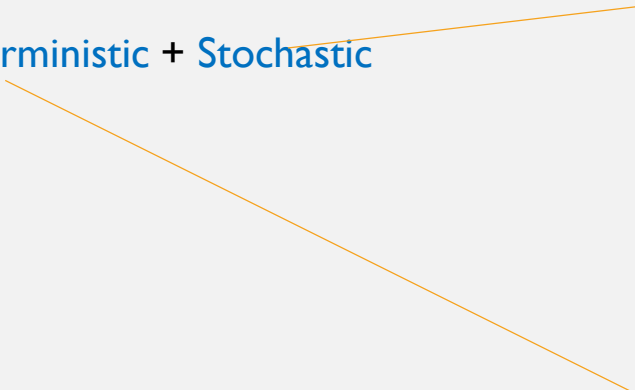
- After you fit a regression model, it is crucial to check the residual plots.
- If the plots display unwanted patterns, you can't trust the regression coefficients and other numeric results.
- Residual plots display the residual values on the y-axis and fitted values, or another variable, on the x-axis.

ESSENTIAL PARTS OF A REGRESSION MODEL

Dependent Variable = (Constant + Independent Variables) + Error

Or:

Dependent Variable = Deterministic + Stochastic



- Stochastic just means unpredictable.
- the difference between the expected value and the observed value.
- series of errors, should look random

- deterministic component is the portion of the variation in the dependent variable that the independent variables explain.
- all of the explanatory power should reside here.

R-SQUARED

- Is a goodness-of-fit measure for linear regression models.
- Indicates the % of the variance in the dependent variable that the independent variables explain collectively.
- R-squared measures the strength of the relationship between your model and the dependent variable on a convenient 0 – 100% scale.
 - the R^2 is always going to be between $-\infty$ and 1
- *Small R-squared values are not always a problem, and high R-squared values are not necessarily good!*

R² - INTERPRETATION

$$SST = \sum (y - \bar{y})^2$$

$$SSM = \sum (\hat{y} - \bar{y})^2$$

$$SSE = \sum (y - \hat{y})^2$$

$$R^2 = \frac{SSM}{SST}$$

R^2 will increase as we increase the number of features or complexity

Refer to python code
[ML-LINREG-01-r2-adj-r2](#)

EXERCISE

X	Y
1	8
2	8
3	8
4	8
5	8
6	8
7	8
8	8
9	8
10	8

R² of this linear regression line is?

ADJUSTED R²

$$R^2_{\text{adjusted}} = 1 - \frac{(1 - R^2)(N - 1)}{N - p - 1}$$

where

R^2 = sample R-square

p = Number of predictors

N = Total sample size.

INTERPRET P-VALUES AND COEFFICIENTS

- **P-values** and **coefficients** in regression analysis work together to tell which relationships in the model are statistically significant and the nature of those relationships.
- The coefficients describe the mathematical relationship between each independent variable and the dependent variable.
- The p-values for the coefficients indicate whether these relationships are statistically significant.
- *After fitting a regression model, check the residual plots first to be sure that we have unbiased estimates. After that, it's time to interpret the statistical output.*

BASICALLY ...

- Regression analysis is a form of inferential statistics.
- The **p-values** help determine whether the relationships observed in the sample also exist in the larger population.
- The p-value for each independent variable tests the null hypothesis that the **variable has no correlation** with the dependent variable. (no effect)

p – value

$$= \left[\begin{array}{c} \text{if less than 0.05, then reject the Null hypothesis, that variables} \\ \text{have no effect. Keep those variables in the model} \\ \hline \text{if more than 0.05 then ACCEPT the Null hypothesis, that the variables} \\ \text{have no effect. Remove these features} \end{array} \right]$$

SUMMARY

Acronym	Residual Operation ?	Robust To Outliers ?	Key Observations (pros)	Key Observations (cons)
MAE Mean Absolute Error	Absolute Value	Yes	• Mean magnitude of the residuals • larger errors will contribute linearly to the overall error. • small MAE suggests the model is great at prediction	• does not indicate if the model under or overshoots actual data). • large MAE suggests that your model may have trouble in certain areas

SUMMARY

Acronym	Residual Operation ?	Robust To Outliers ?	Key Observations (pros)	Key Observations (cons)
MSE Mean Squared Error	Square	No	Analogous to variance	- error grows quadratically Overestimating badness - Metric worsens with the square term in case of few outliers - even a “perfect” model may have a high MSE in case of noisy data underestimate badness ALL the errors are small

SUMMARY

Acronym	Residual Operation?	Robust To Outliers?	Key Observations (pros)	Key Observations (cons)
RMSE Root Mean Squared Error	Square	No	• a close relative of MSE, can compare models using MSE • Analogous to SD	• affected by the outliers

SUMMARY

Acronym	Residual Operation ?	Robust To Outliers ?	Key Observations (pros)	Key Observations (cons)
MAPE Mean Absolute Percentage Error	Absolute Value	Yes	• percentage equivalent of MAE	• undefined for data points where the value is 0 • can grow unexpectedly large if the actual values are exceptionally small themselves • biased towards predictions that are systematically less than the actual values

SUMMARY

Acronym	Full Name	Residual Operation?	Robust To Outliers?	Key Observations (pros)	Key Observations (cons)
MPE	Mean Percentage Error	N/A	Yes	• more negative or positive errors, this bias will show up in the MPE • Tells if our model systematically underestimates (more negative error) or overestimates (positive error).	Since positive and negative errors cancel out, cannot make any statements about how well the model predictions perform overall

PARAMETERS

```
sklearn.linear_model.LinearRegression(fit_intercept=True,  
                                     normalize=False,  
                                     copy_X=True,  
                                     n_jobs=None)
```

Parameters

fit_intercept	boolean, optional, default True whether to calculate the intercept for this model. If set to False, no intercept will be used in calculations
normalize	boolean, optional, default False If True, the regressors X will be normalized before regression by subtracting the mean and dividing by the l2-norm. If you wish to standardize, please use sklearn.preprocessing.StandardScaler before calling fit on an estimator with <code>normalize=False</code> .

IMPORTANT POINTS:

- There **must be linear relationship** between independent and dependent variables
- Multiple regression suffers from **multicollinearity**, **autocorrelation**, **heteroskedasticity**.
- Linear Regression is very **sensitive to Outliers**. It can terribly affect the regression line and eventually the forecasted values.
- **Multicollinearity** can increase the variance of the coefficient estimates and make the estimates very sensitive to minor changes in the model. The result is that the coefficient estimates are unstable
- In case of multiple independent variables, go with *forward selection, backward elimination and step wise approach for selection of most significant independent variables*.

WHEN TO USE LIN REG

$$Y_i = \alpha + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_p X_{ip} + \varepsilon_i$$

where,

$\beta_1, \beta_2, \dots, \beta_p$
 $X_{i1}, X_{i2}, \dots, X_{ip}$

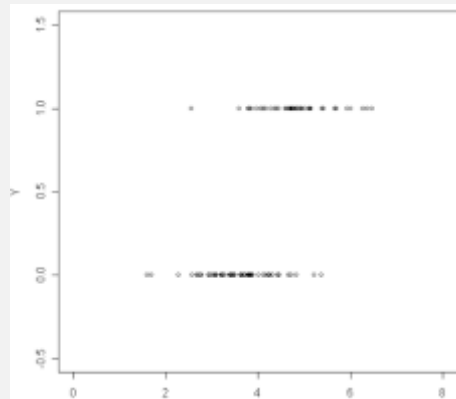
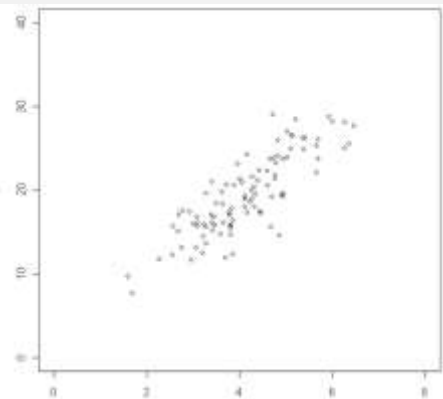
α

ε_i

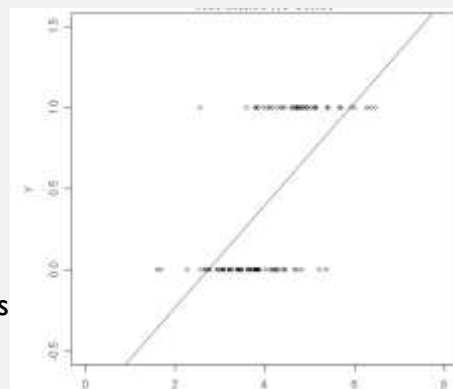
are the regression slope coefficients,
are the corresponding values of the X-variables,
is the intercept, and
is the random error term.

the error term Epsilon has a normal distribution with mean 0 and standard deviation

This means the output y , must be continuous variable (i.e., one that takes values on an interval), not a binary variable



- The least-squares regression line gives predictions that make no sense.
- For example, for an X value of 8, the least-square regression line predicts that Y will be above 1.5.
- But Y can only take on values of 0 and 1.



ASSUMPTIONS

Criteria

Number of observations	the observation-to-Independent Variables (IVs) ratio should ideally be 20:1; that is 20 cases for every IV in the model. For 10 IV, total sample size min 200
Accuracy of data	Pandas (describe method) at least check the minimum and maximum value for each variable to ensure that all values for each variable are "valid."
Missing data	Note: regression analysis in some tools, drops all observations that have a missing value for any one of the variables used in the model - Consider imputation (In case of few values being missing)
Outliers	• An outlier is often operationally defined as a value that is at least 3 SD above or below the mean. • delete those cases. • recode the value

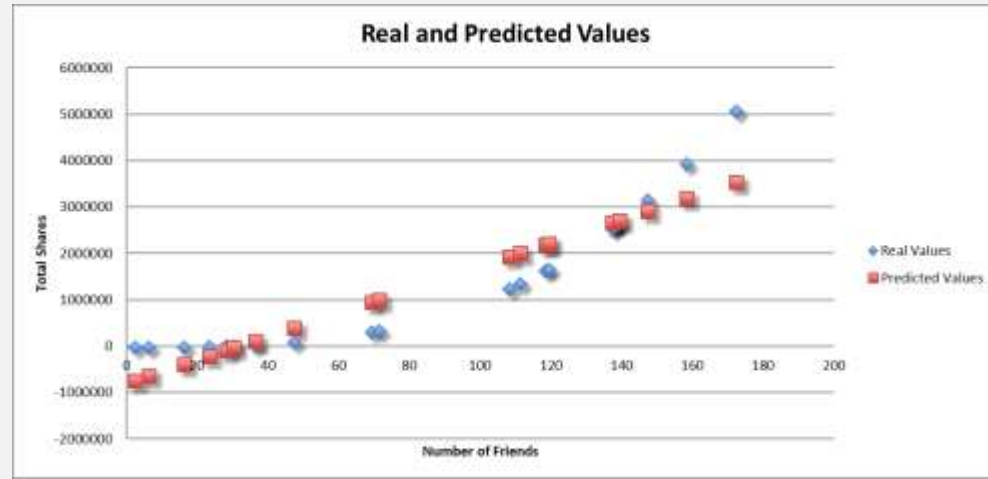
ASSUMPTIONS

Criteria

Normality	construct histograms and "look" at the data to see its distribution.
Linearity	bivariate scatterplot
Multicollinearity	• Multicollinearity is a condition in which the IVs are very highly correlated (.90 or greater) • Ordinary Least square method cannot be that effective. • deleting one of the two variables (after careful analysis) • Statistically, regression coefficients is calculated through matrix inversion. if multicollinearity exists the inversion is unstable.
Transformations	• one or more variables are not normally distributed, you might want to transform them • harder to interpret the analysis

WHEN, WHY AND HOW SHOULD USE LINEAR REGRESSION

- The relationship between the variables is linear.
 - **variance** in the **residuals** (the difference in the real and predicted values) is more or less **constant**.
- The residuals are independent, meaning the residuals are distributed randomly and not influenced by the residuals in previous observations.
- The residuals are normally distributed. This assumption means the

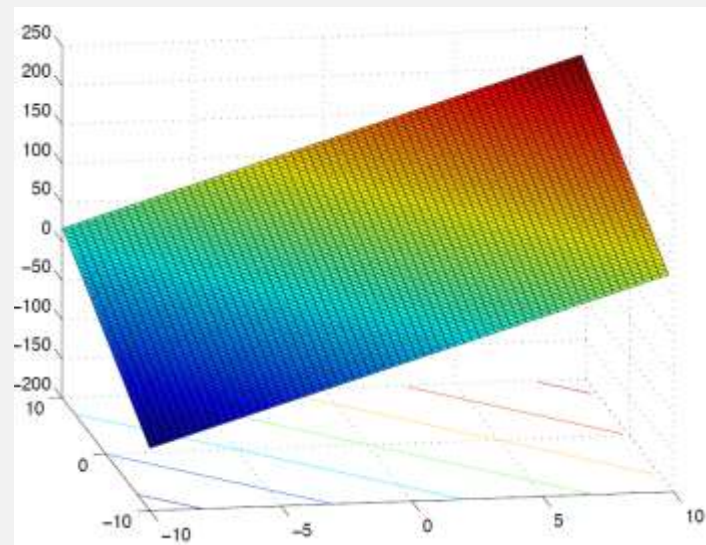


USE CASES

- Sales of a product; pricing, performance, and risk parameters
- Generating insights on consumer behavior, profitability, and other business factors
- Evaluation of trends; making estimates, and forecasts
- Determining marketing effectiveness, pricing, and promotions on sales of a product
- Assessment of risk in financial services and insurance domain
- Studying engine performance from test data in automobiles
- Calculating causal relationships between parameters in biological systems
- Conducting market research studies and customer survey results analysis
- Astronomical data analysis
- Predicting house prices with the increase in sizes of houses
- stock trading, video games, sports betting, and flight time prediction.

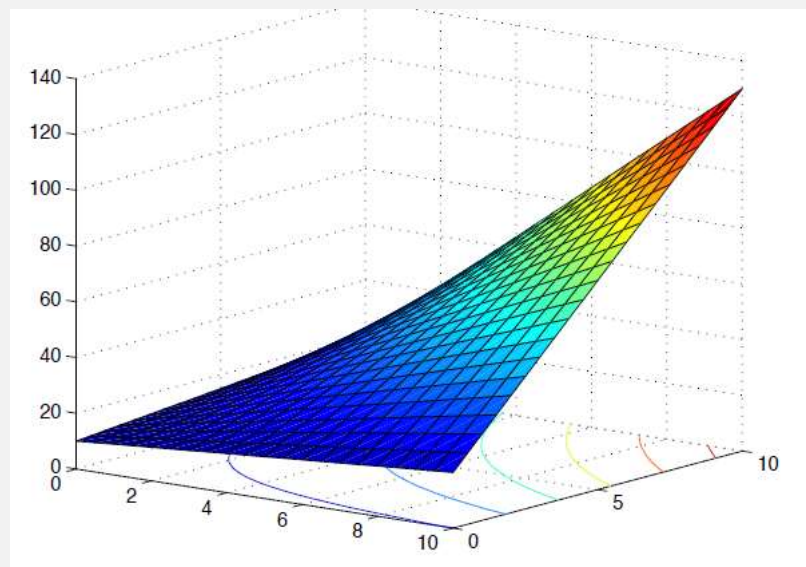
GEOMETRIC INTUITION

- Models with 2 predictor variables (say x_1 and x_2) and a response variable y can be understood as a two-dimensional surface in space.
- The observations are points in space and the surface is “fitted” to best approximate the observations.
- Simplest multiple regression model for 2 predictor variables is $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$
- The surface that corresponds to the model $y = 50 + 10x_1 + 7x_2$ looks like this.
- It is a plane in R^3 with different slopes in x_1 and x_2



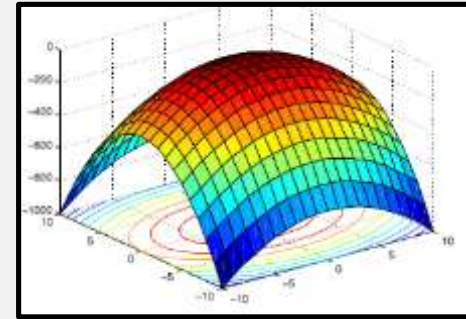
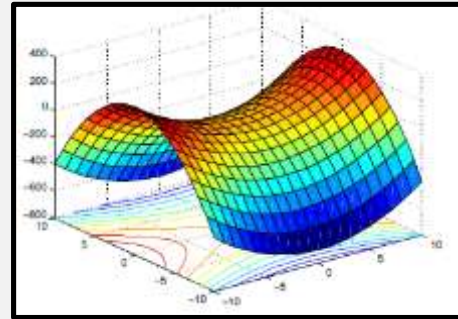
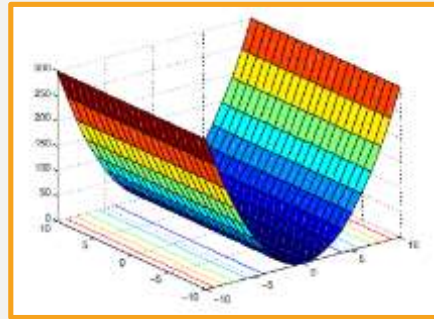
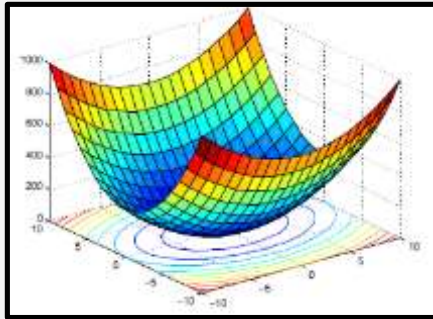
GEOMETRIC INTUITION

- For a simple linear model with two predictor variables and an interaction term, the surface is no longer flat but curved.
- $y = 10 + x_1 + x_2 + x_1x_2$



GEOMETRIC INTUITION

- **Polynomial regression** models with 2 predictor variables and **interaction terms** are quadratic forms.
- Their surfaces can have many different shapes depending on the values of the model parameters with the contour lines being either **parallel lines, parabolas or ellipses**.
- $y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_{11} X_1^2 + \beta_{22} X_2^2 + \beta_{12} X_1 X_2 + \varepsilon$



DECISION TREES



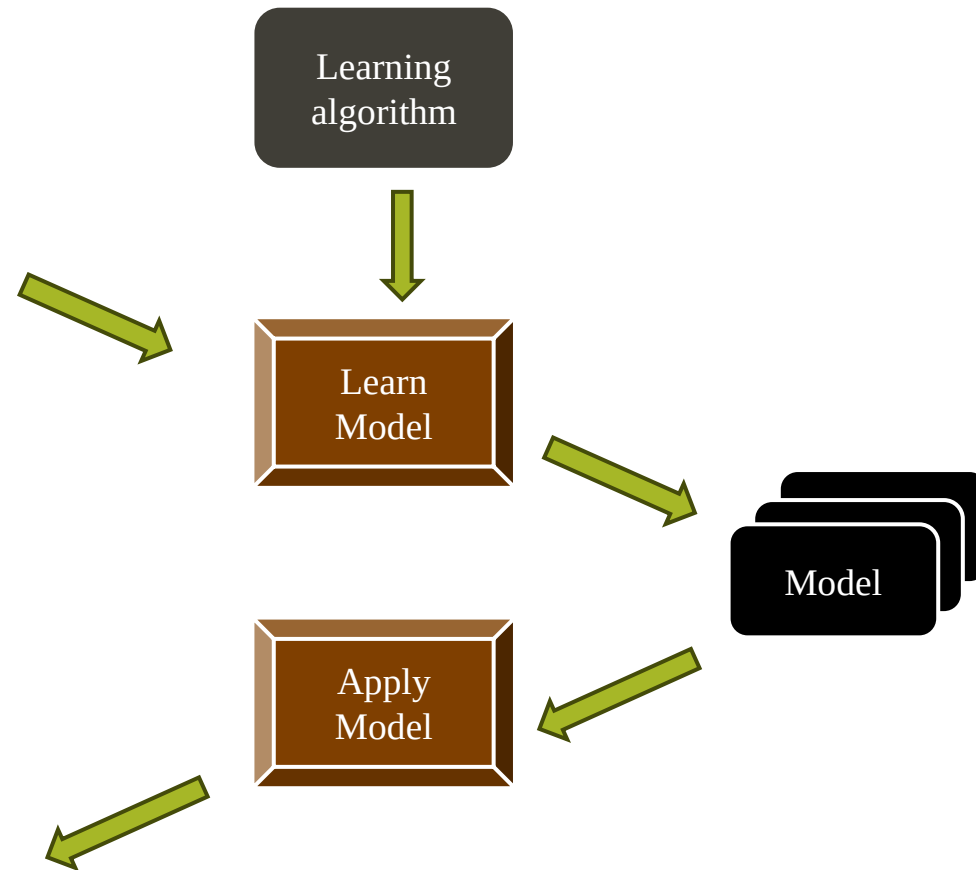
Classification Task (Decision Trees)

Gender	Occupation	Suggestion
F	Student	PUBG
F	Programmer	Github
M	Programmer	Whatsapp
F	Programmer	Github
M	Student	PUBG
M	Student	PUBG

Training set

Gender	Occupation	Suggestion
F	Student	PUBG
M	Student	PUBG

Test set

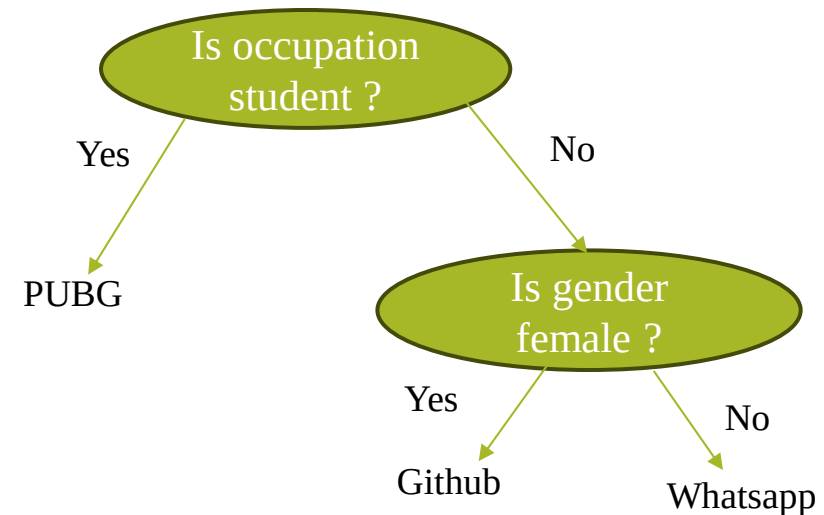


Decision Trees

Example Dataset

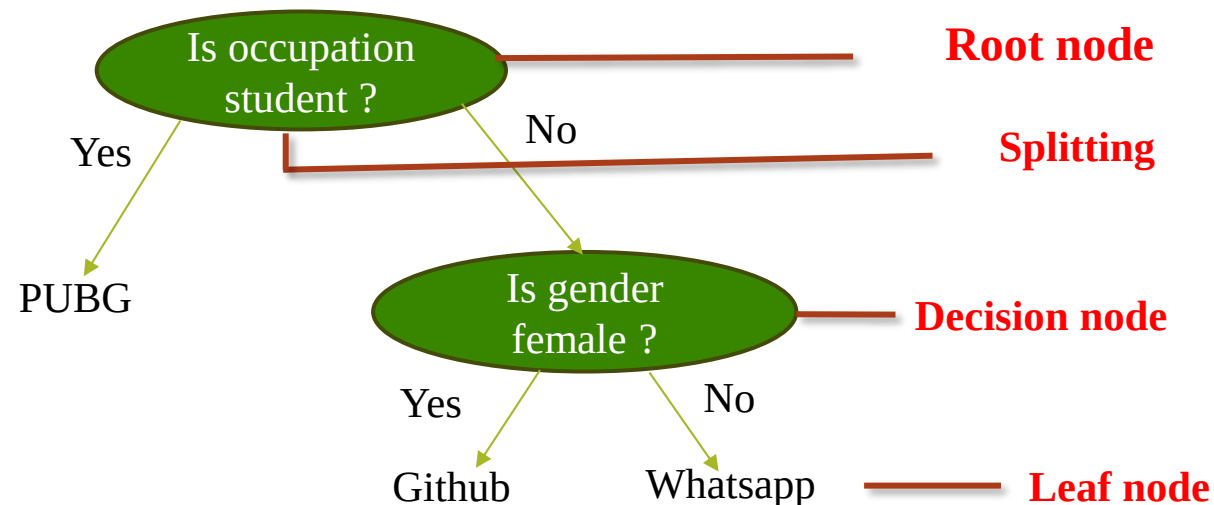
Gender	Occupation	Suggestion
F	Student	PUBG
F	Programmer	Github
M	Programmer	Whatsapp
F	Programmer	Github
M	Student	PUBG
M	Student	PUBG

```
If occupation == student
    print (PUBG)
Else
    If gender == female
        print(Github)
    Else
        print (Whatsapp)
```

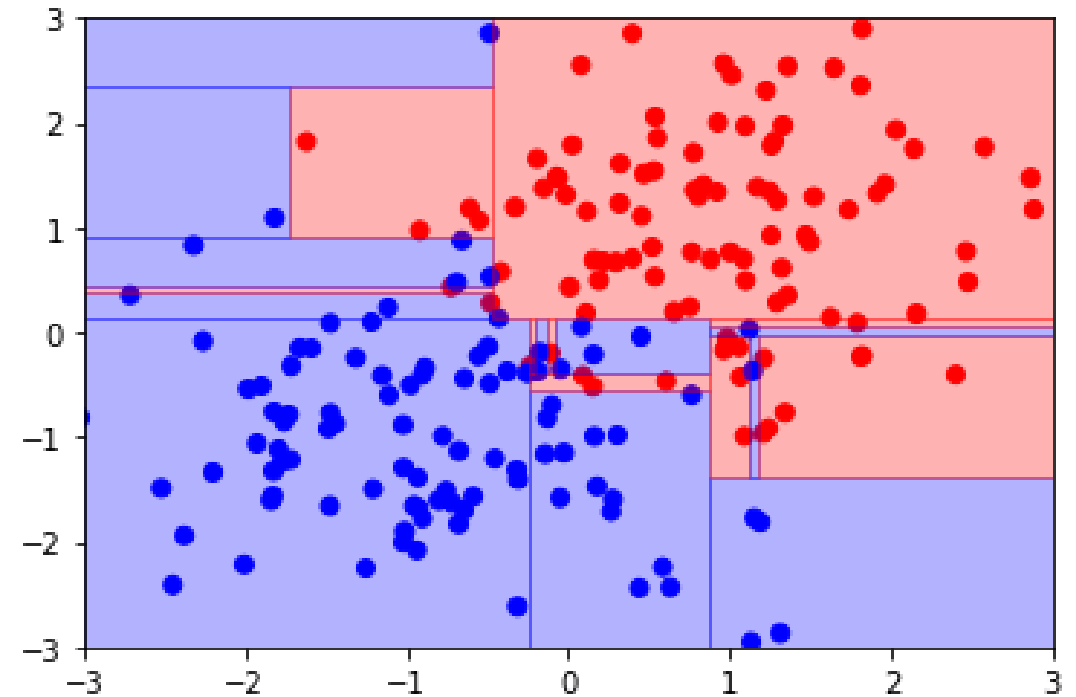
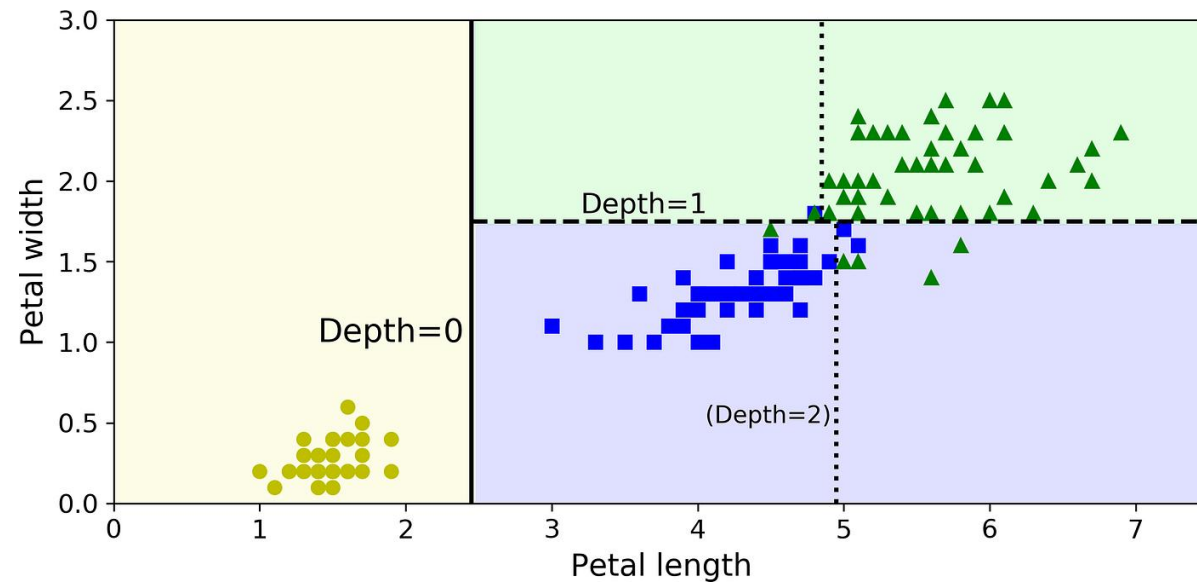


Decision Trees (Pseudo code)

- ✓ Begin with your **training dataset**, which should have some feature variables and classification or regression output.
- ✓ Determine the **“best feature”** in the dataset to split the data on; more on how we define “best feature later”
- ✓ Split the data into subsets that contain the correct values for this best feature. This splitting basically defined a node on the tree i.e each node is a splitting point based on a certain feature from our data.



Decision Trees (Geometric Intuition)



Decision Trees

Some unanswered questions ?

- I. How to decide which column should be considered as root node?
- II. How to select subsequent decision nodes?

Advantages:

- I. Intuitive and easy to understand
- II. Minimal data preparation is required
- III. The cost of using the tree of inference is logarithmic if the number of data points used to train the tree

Disadvantages:

- I. Overfitting
- II. Prone to errors for imbalanced datasets.

Decision Trees

Popular Decision Tree Algorithms

1. **ID3**: Uses **information gain** to split data
2. **CART**: It minimizes **Gini impurity** for classification and **MSE** for regression with **pruning technique** to prevent overfitting.

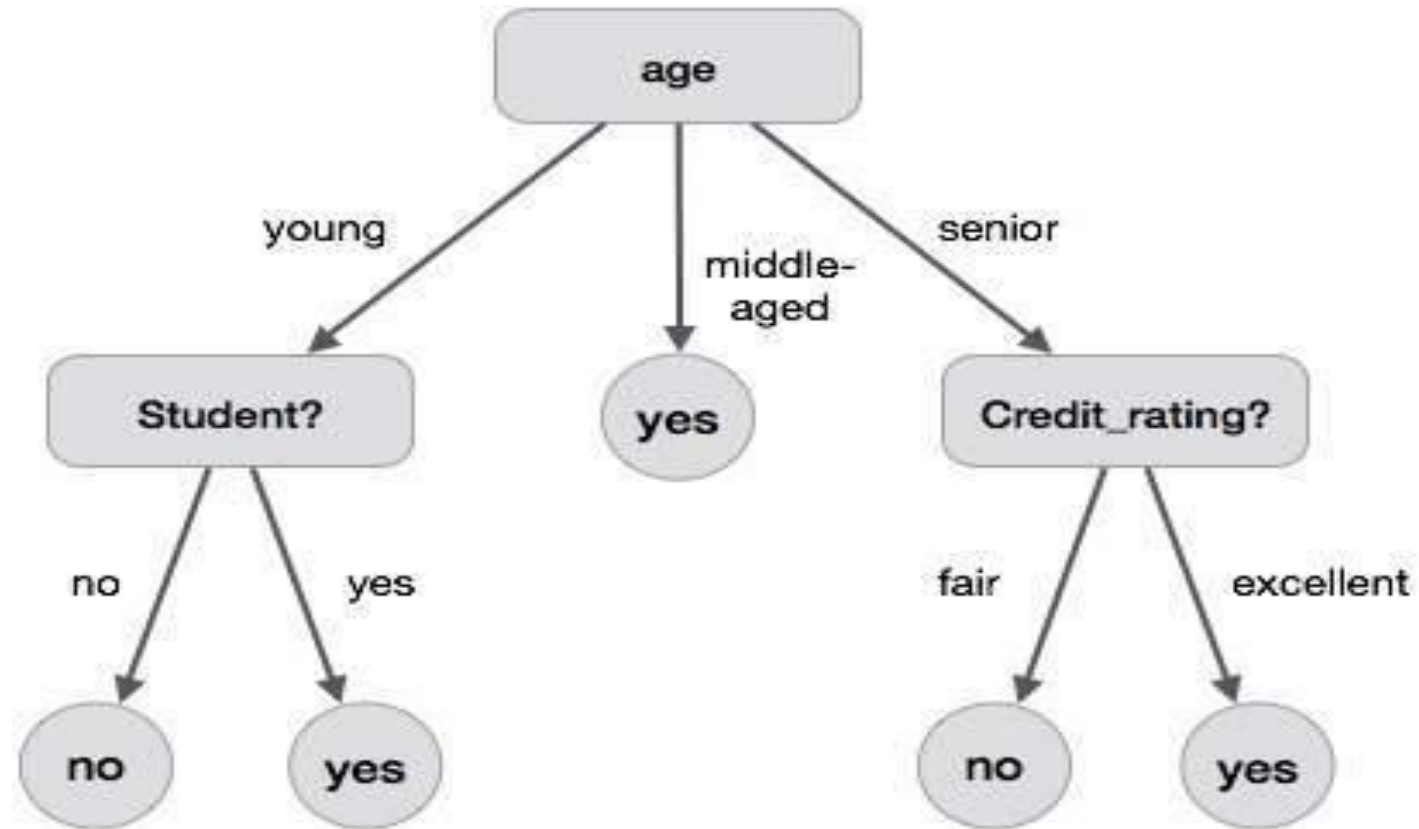
ID3

- ✓ Information Gain, is a metric used to train Decision Trees.
- ✓ Specifically, this metric measures the quality of a split.
- ✓ The information gain is based on the decrease in entropy after a data-set is split on an attribute. Constructing a decision tree is all about finding attribute that returns the high information gain.

Decision Trees

- Decision tree is a simple but powerful learning Paradigm.
- It is a type of classification algorithm for supervised learning.
- A decision tree is a tree in which each branch node represents a choice between a number of alternatives and each leaf node represents a decision.
- A node with outgoing edges is called an internal or test node. All other nodes are called leaves (also known as terminal or decision nodes).

Decision Trees



Construction of decision tree

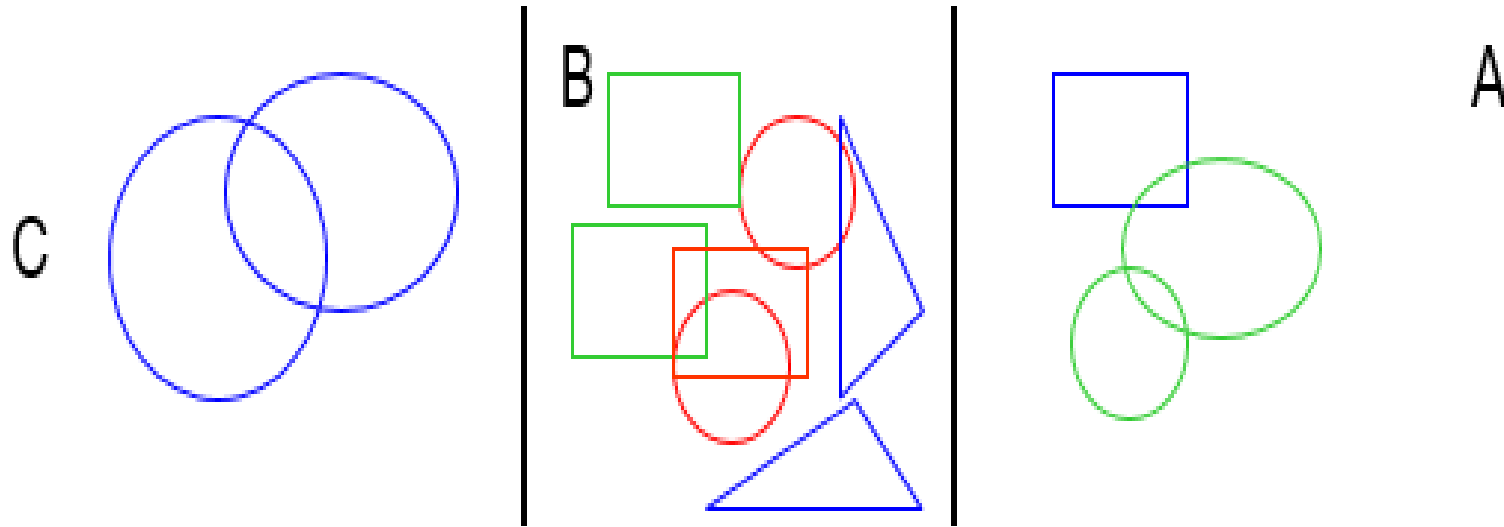
- 1) First test all attributes and select the one that will function as the best root.
- 2) Break-up the training set into subsets based on the branches of the root node.
- 3) Test the remaining attributes to see which ones fit best underneath the branches of the root node.
- 4) Continue this process for all other branches until

Decision Trees

- I. All the examples of a subset are of one type.
- II. There are no examples left.
- III. There are no more attributes left.

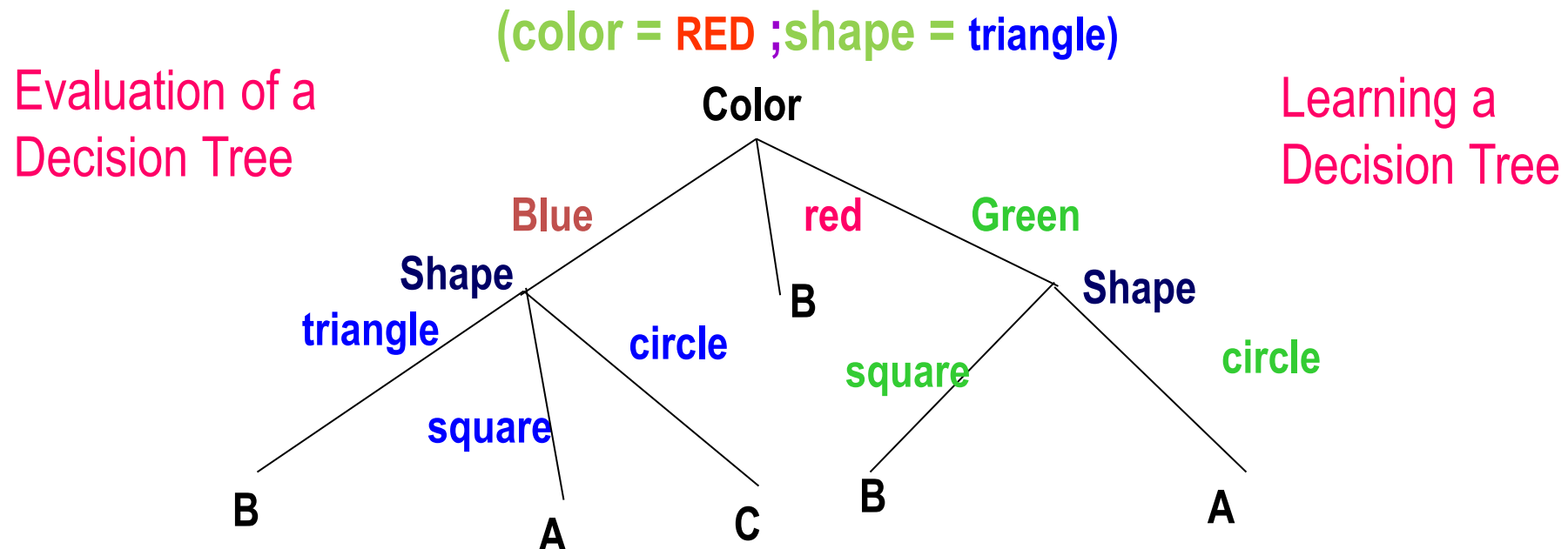
Decision Tree Example

- Given a collection of shape and colour example, learn a decision tree that represents it. Use this representation to classify new examples.



Decision Trees: The Representation

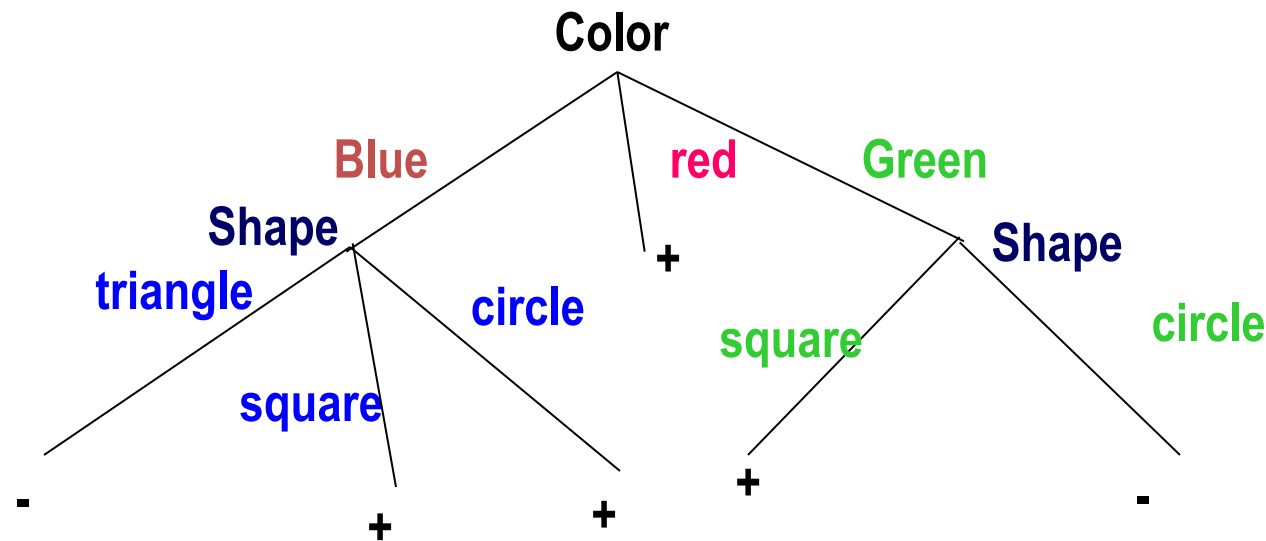
- Decision Trees are classifiers for instances represented as features vectors. (color = ;shape = ;label =)
- Nodes are tests for feature values;
- There is one branch for each value of the feature
- Leaves specify the categories (labels)
- Can categorize instances into multiple disjoint categories – multi-class



Binary classified Decision Trees

They can represent **any Boolean function**

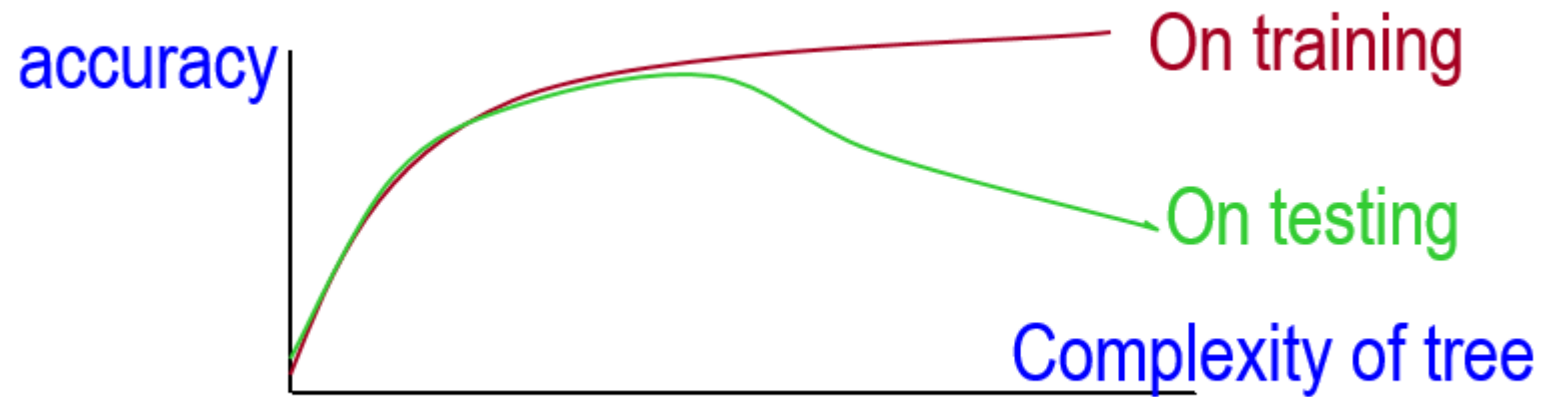
- Can be rewritten as rules in Disjunctive Normal Form (DNF)
- $\text{green} \wedge \text{square} \rightarrow \text{positive}$
- $\text{blue} \wedge \text{circle} \rightarrow \text{positive}$
- $\text{blue} \wedge \text{square} \rightarrow \text{positive}$
- The disjunction of these rules is equivalent to the Decision Tree



Limitation of Decision Trees

- Learning a tree that classifies the training data perfectly may not lead to the tree with the **best generalization performance**.
 - There may be **noise** in the training data the tree is fitting
 - The algorithm might be making decisions based on **very little data**.
- A hypothesis **h** is said to overfit the training data if there is another hypothesis, **h'** , such that ' **h** ' has smaller error than **h'** on the training data but **h** has larger error on the test data than **h'** .

Decision Trees



Evaluation Methods for Decision Trees

- Two basic approaches
 - **Pre-pruning:** Stop growing the tree at some point during construction when it is determined that there is not enough data to make reliable choices.
 - **Post-pruning:** Grow the full tree and then remove nodes that seem not to have sufficient evidence.
- Methods for evaluating subtrees to prune:
 - **Cross-validation:** Reserve hold-out set to evaluate utility.
 - **Statistical testing:** Test if the observed regularity can be dismissed as likely to be occur by chance.
 - **Minimum Description Length:** Is the additional complexity of the hypothesis smaller than remembering the exceptions ?

ID3 (Iterative Dichotomiser 3)

- Invented by J.Ross Quinlan in 1975.
- Used to generate a decision tree from a given data set by employing a top-down, greedy search, to test each attribute at every node of the tree.
- The resulting tree is used to classify future samples.

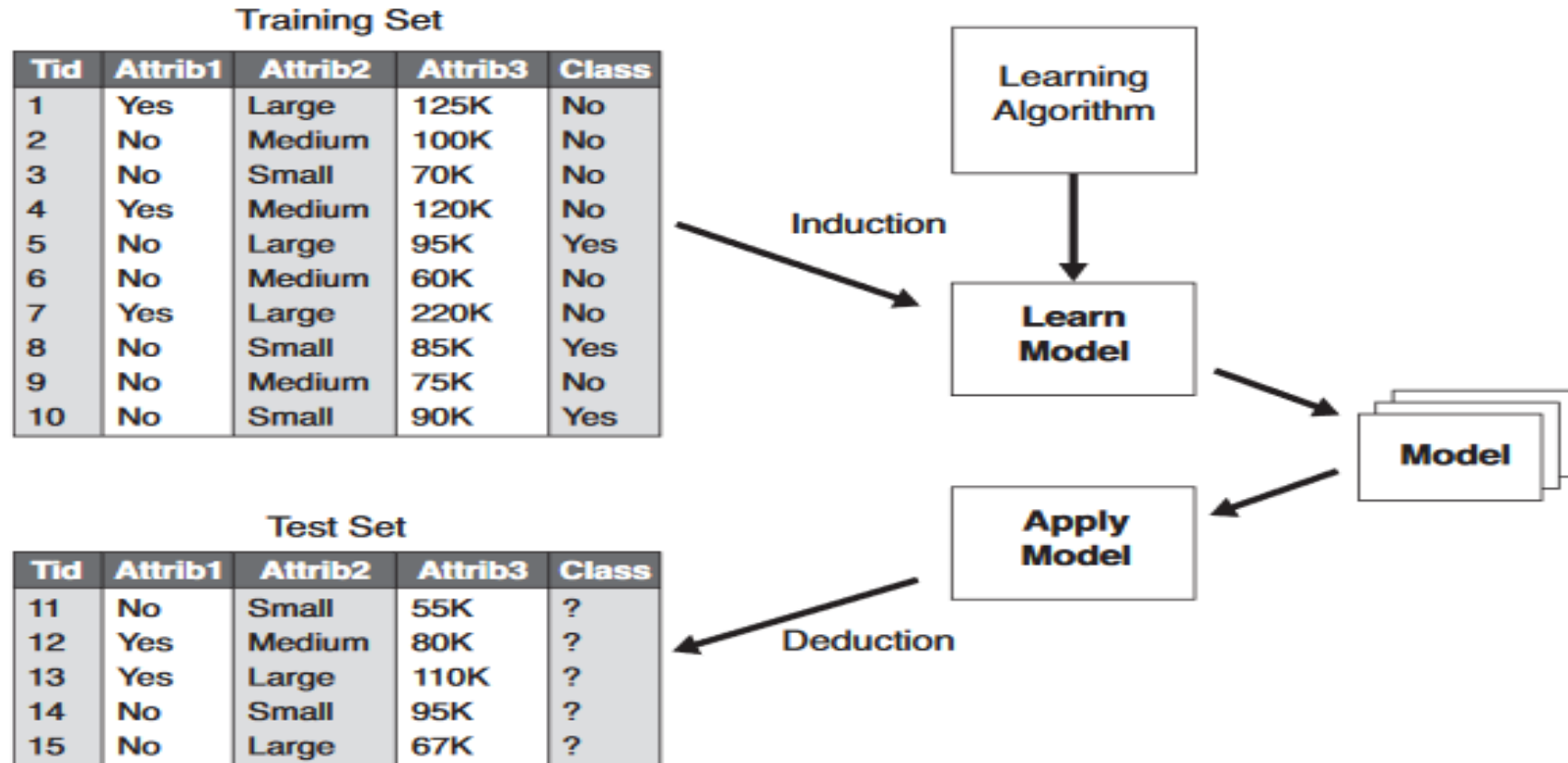
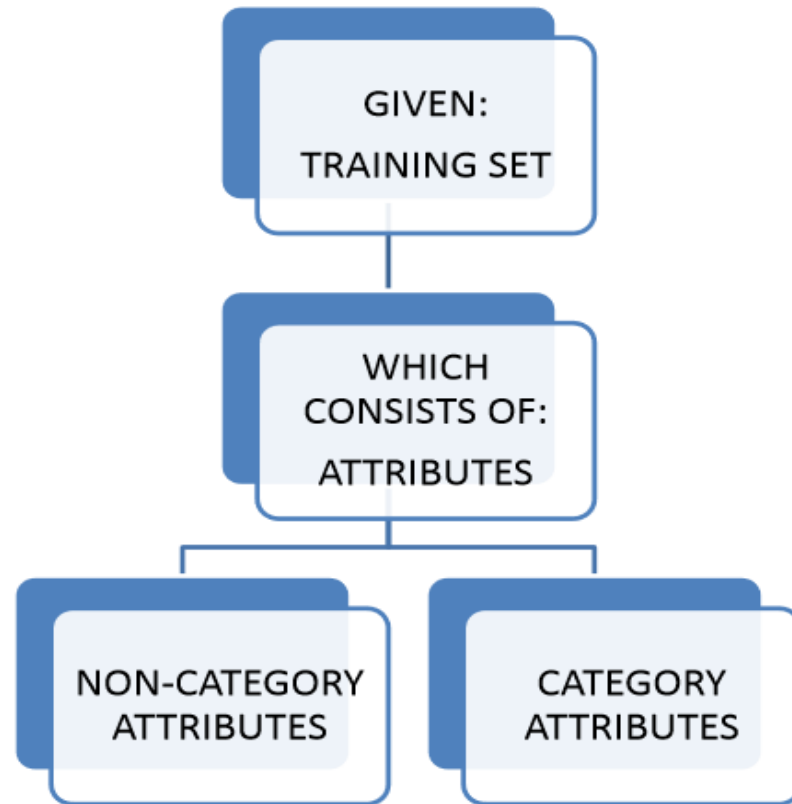


Figure 4.3. General approach for building a classification model.



Training Set

Attributes				Classes
Gender	Car Ownership	Travel Cost	Income Level	Transportation
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	1	Cheap	Medium	Train
Female	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Male	0	Standard	Medium	Train
Female	1	Standard	Medium	Train
Female	1	Expensive	High	Car
Male	2	Expensive	Medium	Car
Female	2	Expensive	High	Car

Algorithm

- Calculate the **Entropy** of every attribute using the data set.
- Split the set into subsets using the attribute for which entropy is minimum (or, equivalently, information gain is maximum).
- Make a decision tree node containing that attribute.
- Recurse on subsets using remaining attributes.

Entropy

- In order to define **Information Gain** precisely, we need to discuss **Entropy** first.
- A formula to calculate the homogeneity of a sample.
- A completely homogeneous sample has entropy of 0 (leaf node).
- An equally divided sample has entropy of 1.
- The formula for entropy is:

$$\text{Entropy}(S) = -\sum p(I) \log_2 p(I)$$

- where $p(I)$ is the proportion of S belonging to class I . $\sum$ is over total outcomes.

Entropy

- Example 1

If S is a collection of 14 examples with 9 YES and 5 NO examples then

$$\begin{aligned}\text{Entropy}(S) &= - (9/14) \text{Log}_2 (9/14) - (5/14) \text{Log}_2 (5/14) \\ &= 0.940\end{aligned}$$

Information Gain

- The information gain is based on the decrease in entropy after a dataset is split on an attribute.
- The formula for calculating information gain is:

$$\text{Gain}(S, A) = \text{Entropy}(S) - ((|S_v| / |S|) * \text{Entropy}(S_v))$$

Where, S_v = subset of S for which attribute A has value v

- $|S_v|$ = number of elements in S_v
- $|S|$ = number of elements in S

Procedure

- First the entropy of the total dataset is calculated.
- The dataset is then split on the different attributes.
- The entropy for each branch is calculated.
- Then it is added proportionally, to get total entropy for the split.
- The resulting entropy is subtracted from the entropy before the split.
- The result is the Information Gain, or decrease in entropy.
- The attribute that yields the largest IG is chosen for the decision node.

Our Problem :

Name	Gender	Car Ownership	Travel Cost	Income Level	Transportation
Abhi	Male	1	Standard	High	?
Pavi	Male	0	Cheap	Medium	?
Ammu	Female	1	Cheap	High	?

Training Set

Attributes				Classes
Gender	Car Ownership	Travel Cost	Income Level	Transportation
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	1	Cheap	Medium	Train
Female	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Male	0	Standard	Medium	Train
Female	1	Standard	Medium	Train
Female	1	Expensive	High	Car
Male	2	Expensive	Medium	Car
Female	2	Expensive	High	Car

Calculate the entropy of the total dataset

- First compute the Entropy of given training set.

Probability

Bus : 4/10 = 0.4

Train: 3/10 = 0.3

Car: 3/10 = 0.3

$$E(S) = -P(I) \log_2 P(I)$$

$$E(S) = -(0.4) \log_2 (0.4) - (0.3) \log_2 (0.3) - (0.3) \log_2 (0.3) = 1.571$$

Attributes				Classes
Gender	Car Ownership	Travel Cost	Income Level	Transportation
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	1	Cheap	Medium	Train
Female	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Male	0	Standard	Medium	Train
Female	1	Standard	Medium	Train
Female	1	Expensive	High	Car
Male	2	Expensive	Medium	Car
Female	2	Expensive	High	Car



Attributes	Classes
Gender	Transportation
Male	Bus
Male	Bus
Female	Train
Female	Bus
Male	Bus
Male	Train
Female	Train
Female	Car
Male	Car
Female	Car

Split the dataset on 'Gender' attribute

Attributes	Classes
Gender	Transportation
Male	Bus
Male	Bus
Male	Bus
Male	Train
Male	Car

Probability
 Bus : $3/5 = 0.6$
 Train: $1/5 = 0.2$
 Car: $1/5 = 0.2$

$$E(S_v) = -0.6 \log_2 (0.6) - 0.2 \log_2 (0.2) - 0.2 \log_2 (0.2) = 1.522$$

Attributes	Classes
Gender	Transportation
Female	Train
Female	Bus
Female	Train
Female	Car
Female	Car

Probability
 Bus : $1/5 = 0.2$
 Train: $2/5 = 0.4$
 Car: $2/5 = 0.4$

$$E(S_v) = -0.2 \log_2 (0.2) - 0.4 \log_2 (0.4) - 0.4 \log_2 (0.4) = 1.371$$

$$\begin{aligned} \text{Gain}(S,A) &= E(S) - I(S,A) \\ I(S,A) &= 1.522 * (5/10) + 1.371 * (5/10) \end{aligned}$$

$$\begin{aligned} \text{Gain}(S,A) &= 1.571 - 1.447 \\ &= 0.12 \end{aligned}$$

Attributes				Classes
Gender	Car Ownership	Travel Cost	Income Level	Transportation
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	1	Cheap	Medium	Train
Female	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Male	0	Standard	Medium	Train
Female	1	Standard	Medium	Train
Female	1	Expensive	High	Car
Male	2	Expensive	Medium	Car
Female	2	Expensive	High	Car



Attributes	Classes
Car Ownership	Transportation
0	Bus
1	Bus
1	Train
0	Bus
1	Bus
0	Train
1	Train
1	Car
2	Car
2	Car

Split the dataset on 'ownership' attribute

Attributes	Classes
Car Ownership	Transportation
0	Bus
0	Bus
0	Train

Probability
Bus : $2/3 = 0.6$
Train: $1/3 = 0.3$
Car: $0/3 = 0$
Entropy = 0.918

Attributes	Classes
Car Ownership	Transportation
1	Bus
1	Train
1	Bus
1	Train
1	Car

Probability
Bus : $2/5 = 0.4$
Train: $2/5 = 0.4$
Car: $1/5 = 0.2$
Entropy = 1.522

Attributes	Classes
Car Ownership	Transportation
2	Car
2	Car

Probability
Bus : $0/2 = 0$
Train: $0/2 = 0$
Car: $2/2 = 1$
Entropy = 0

$$\begin{aligned} \text{Gain}(S,A) &= E(S) - I(S,A) \\ I(S,A) &= 0.918 * (3/10) + \\ & 1.522 * (5/10) + 0 * (2/10) \end{aligned}$$

$$\begin{aligned} \text{Gain}(S,A) &= 1.571 - 1.0364 \\ &= 0.534 \end{aligned}$$

- If we choose **Travel Cost** as splitting attribute,

-Entropy for Cheap = 0.722

Standard = 0

Expensive = 0

IG = 1.21

- If we choose **Income Level** as splitting attribute,

-Entropy for Low = 0

Medium = 1.459

High = 0

IG = 0.695

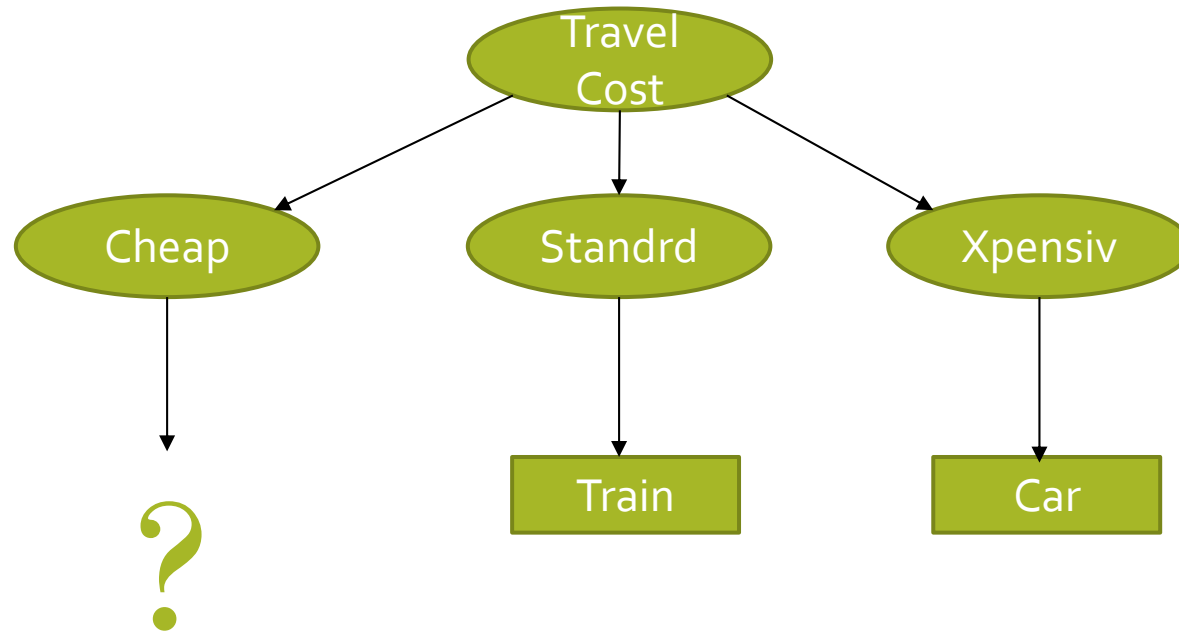
Attribute	Information Gain
Gender	0.125
Car Ownership	0.534
Travel Cost	1.21
Income Level	0.695

Attributes				Classes
Travel Cost	Gender	Car Ownership	Income Level	Transportation
Cheap	Male	0	Low	Bus
Cheap	Male	1	Medium	Bus
Cheap	Female	1	Medium	Train
Cheap	Female	0	Low	Bus
Cheap	Male	1	Medium	Bus

Attributes				Classes
Travel Cost	Gender	Car Ownership	Income Level	Transportation
Standard	Male	0	Medium	Train
Standard	Female	1	Medium	Train

Attributes				Classes
Travel Cost	Gender	Car Ownership	Income Level	Transportation
Expensive	Female	1	High	Car
Expensive	Male	2	Medium	Car
Expensive	Female	2	High	Car

Diagram : Decision Tree



Iteration on Subset of Training Set

Attributes				Classes
Travel Cost	Gender	Car Ownership	Income Level	Transportation
Cheap	Male	0	Low	Bus
Cheap	Male	1	Medium	Bus
Cheap	Female	1	Medium	Train
Cheap	Female	0	Low	Bus
Cheap	Male	1	Medium	Bus

Probability

Bus : $4/5 = 0.8$

Train: $1/5 = 0.2$

Car: $0/5 = 0$

$$E(S) = -P(I) \log_2 P(I)$$

$$E(S) = -(0.8) \log_2 (0.8) - (0.2) \log_2 (0.2) = 0.722$$

- If we choose **Gender** as splitting attribute,

-Entropy for Male = 0

Female = 1

IG = 0.322

- If we choose **Car Ownership** as splitting attribute,

-Entropy for 0 = 0

1 = 0.918

IG = 0.171

- If we choose **Income Level** as splitting attribute,

-Entropy for Low = 0

Medium = 0.918

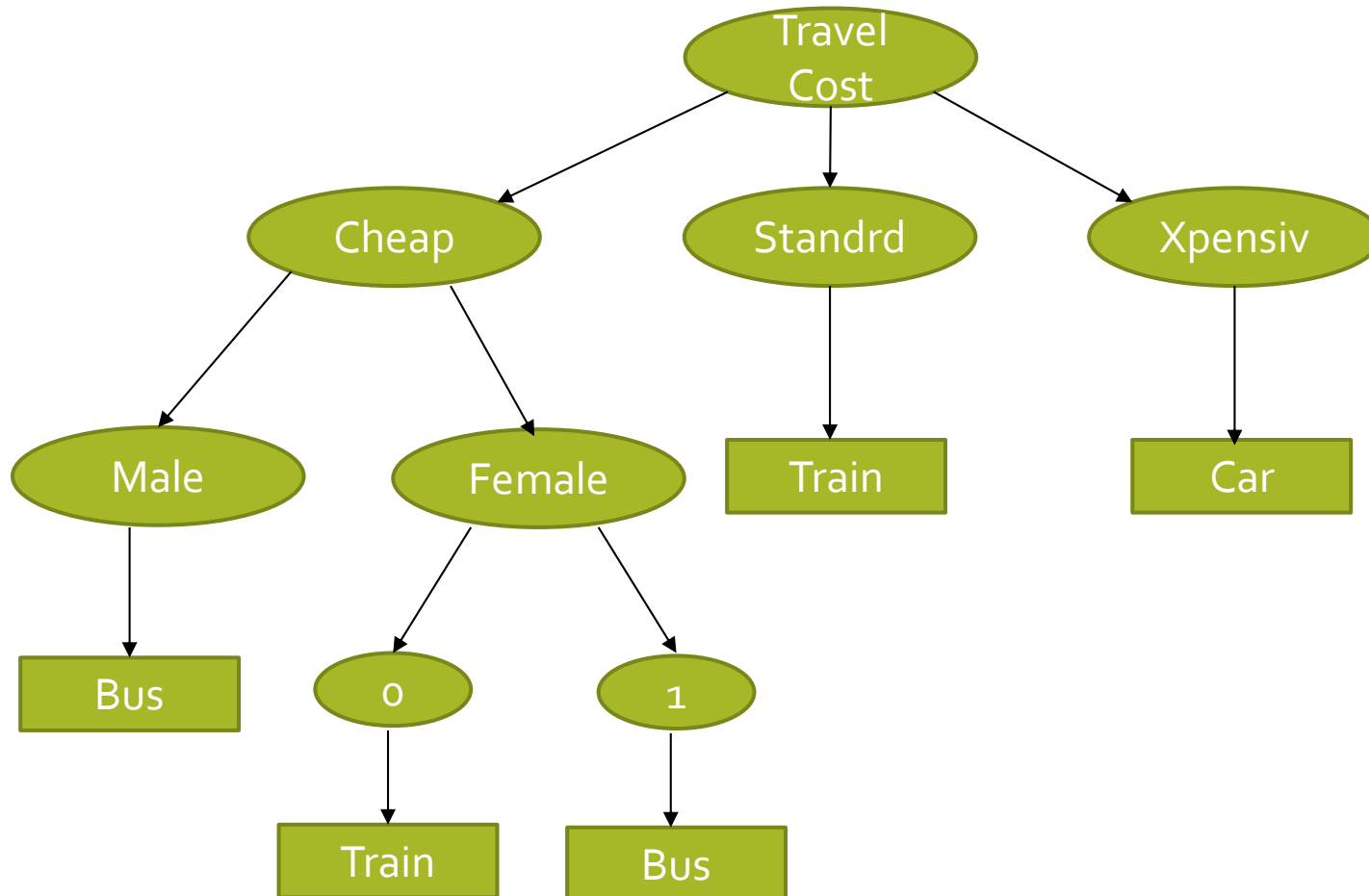
IG = 0.171

Attributes	Information Gain
Gender	0.322
Car Ownership	0.171
Income Level	0.171

Attributes			Classes
Gender	Car Ownership	Income Level	Transportation
Male	0	Low	Bus
Male	1	Medium	Bus
Male	1	Medium	Bus

Attributes			Classes
Gender	Car Ownership	Income Level	Transportation
Female	1	Medium	Train
Female	0	Low	Bus

Diagram : Decision Tree



Solution to Our Problem :

Name	Gender	Car Ownership	Travel Cost	Income Level	Transportat ion
Abhi	Male	1	Standard	High	Train
Pavi	Male	0	Cheap	Medium	Bus
Ammu	Female	1	Cheap	High	Bus

Drawbacks of ID3

- Data may be over-fitted or over-classified, if a small sample is tested.
- Only one attribute at a time is tested for making a decision.
- Classifying continuous data may be computationally expensive.
- Does not handle numeric attributes and missing values.

C4.5 Algorithm

- This algorithm is also invented by J.Ross Quinlan., for generating decision tree.
- It can be treated as an extension of ID3.
- C4.5 is mainly used for classification of data.
- It is called statistical classifier because of its potential for handling noisy data.

Construction

- Base cases are determined.
- For each attribute a , the normalized information gain is found out by splitting on a .
- Let a -best be the attribute with the highest normalized information gain.
- Create a decision node that split on a -best
- Recur on the sub-lists obtained by splitting on a -best, and add those nodes as children of node.

- The new Features include,

- (i) accepts both continuous and discrete features.
- (ii) handles incomplete data points.
- (iii) solves over-fitting problem by bottom-up technique usually known as "pruning".
- (iv) different weights can be applied the features that comprise the training data.

Application based on features

- Continuous attribute categorisation.

discretisation – It partitions the continuous attribute values into discrete set of intervals.

- Handling missing values.

Uses the theory of probability – probability is calculated from the observed frequencies, takes the best probable value after computing the frequencies.

Random Forest

- Random forest is an ensemble classifier which uses many decision trees.
- It builds multiple decision trees and merges them together to get a more accurate and stable prediction.
- It can be used for both classification and regression problems.

Algorithm

- 1) Create a bootstrapped dataset.
- 2) Create a decision tree using the bootstrapped dataset, but only use a random subset of variables(or columns) at each step.
- 3) Go back to step 1 and repeat.

Using a bootstrapped sample and considering only a subset of variables at each step results in a wide variety of trees.

Explanation

- First, the process starts with the randomisation of the data sets (done by bagging)
- Bagging – helps to reduce the variance, helps in effective classification.
- RF contains many classification trees.
- Bagging – Bootstrapping the data plus using the aggregate to make a decision.
- Out-of-Bag-Dataset – The dataset formed by using unchosen tuples from the training set during the process of bootstrapping.

Gini Index

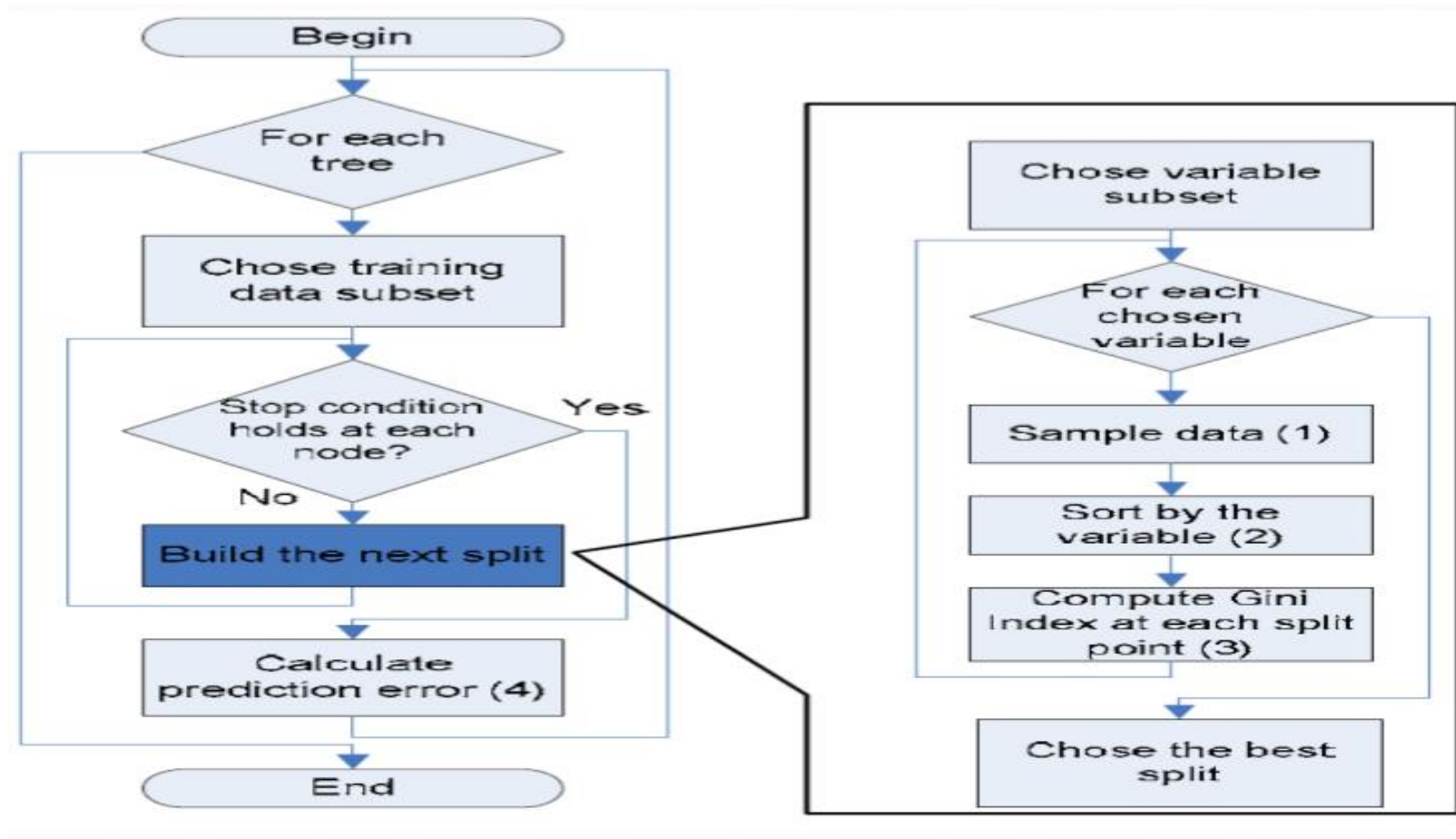
- Random Forest uses the gini index taken from the CART learning system to construct decision trees. The gini index of node impurity is the measure most commonly chosen for classification-type problems. If a dataset T contains examples from n classes.
- gini index, $Gini(T)$ is defined as:

$$Gini(T) = 1 - \sum_{j=1}^n P_j^2$$

where p_j is the relative frequency of class j in T

- If a dataset T is split into two subsets $T1$ and $T2$ with sizes $N1$ and $N2$ respectively, the gini index of the split data contains examples from n classes, the gini index (T) is defined as:
- $$\text{Gini}_{\text{split}}(T) = N_1/N \text{ gini}(T1) + N_2/N \text{ gini}(T2)$$

Flow Chart of Algorithm



Advantages

- It produces a highly accurate classifier and learning is fast
- It runs efficiently on large data bases.
- It can handle thousands of input variables without variable deletion.
- It computes proximities between pairs of cases that can be used in clustering, locating outliers or (by scaling) give interesting views of the data.
- It offers an experimental method for detecting variable interactions.

• Assignment

- Practice decision tree for Continuous data using Information gain and gini impurity.
- Ensemble Learning
- Bagging and Boosting
- Random Forest